Bounds on Codes

We now look at the problem of determining how the parameters n, k, δ of a linear code are constrained. Given values for two of these three parameters, what range of values is possible for the third? This problem has not been solved in general, although it has been settled for many small values of n, k and δ.

5.1.1 Lemma [Spoiling Lemma] If there exists a linear (n, k, δ) code, then there exists a linear (n, k, d) code for $d = 1, 2, \ldots, \delta$.

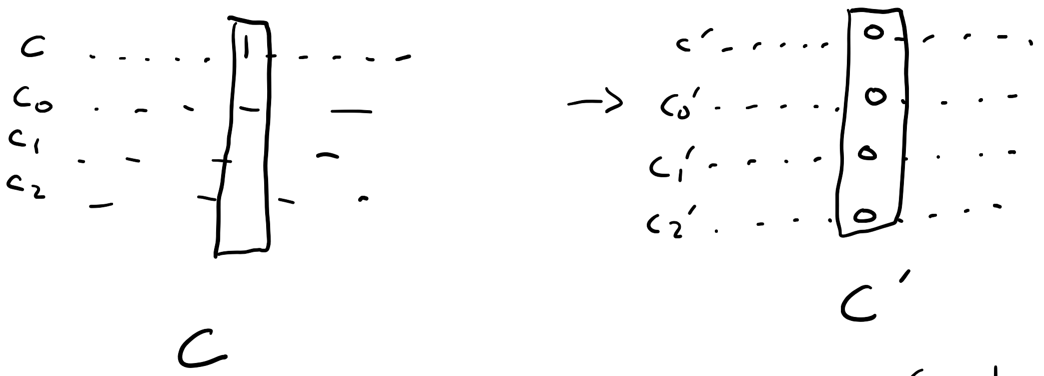
Proof sketch

The case $\delta = 1$ is trivial,

Assume $\delta > 1$, and suppose $\exists$ a linear (n, k, δ) code C.

$\exists\, c \in C$ s.t $\|c\| = \delta$

Pick a position in which c has a 1 and make this position 0 in all the codewords of C, to create a new code C'.



Prove that C' is a linear $(n, k, \delta-1)$ code.

Repeat the process to prove the lemma.

C' has length n

C' is linear

C' has dimension k

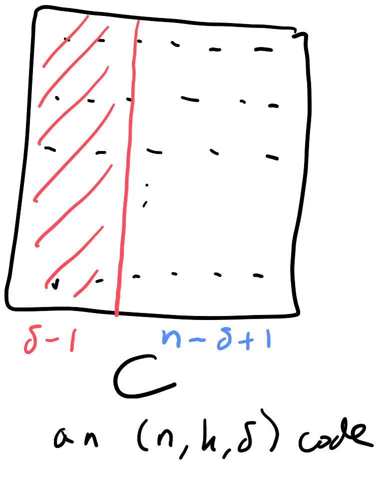
$\delta_{c'} = \delta_c - 1$

5.1.2 Theorem [Singleton Bound] In any code (linear or not), $|C| \leq 2^{n-\delta+1}$. In particular, if $|C| = 2^k$, then

$$n \geq k + \delta - 1. \tag{5.1.1}$$

Proof sketch

Delete the first $\delta-1$ bits of each codeword.



Since C has distance δ the resulting words of length $n-\delta+1$ will be distinct.

There are only $2^{n-\delta+1}$ words of length $n-\delta+1$ so

$$|C| \leq 2^{n-\delta+1}.$$

5.2. The Hamming Bound

5.2.1 Definition Let $v \in \mathbb{F}_2^n$ and let $r \geq 0$ be an integer. The (closed) *ball* of radius r centred at v is the set

$$B_r(v) = \{w \in \mathbb{F}_2^n \mid d(v, w) \leq r\}.$$

Example

$B_2(1000)$

1000

5.2.2 Lemma For any $r \leq n$ and $v \in \mathbb{F}_2^n$

$$|B_r(v)| = \binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{r}.$$

words of length n that differ from v in i places is $\binom{n}{i}$.

5.2.3 Theorem [Hamming Bound] If C is a code of length n (linear or not) and distance δ then

$$|C| \leq \frac{2^n}{\binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{d}} \qquad \text{where } d = \left\lfloor \frac{\delta - 1}{2} \right\rfloor.$$

Every two codewords are at distance $\geq \delta$ and $d = \lfloor \frac{\delta-1}{2} \rfloor$ so the balls $B_d(c)$ do not overlap for any codewords.

So $|C|\left(\binom{n}{0}+\binom{n}{1}+\cdots+\binom{n}{d}\right) \leq$ total # of words of length n $= 2^n$

<u>Exercise</u> Determine the lower bounds on the length of a 2-error correcting code with 64 codewords, given by the Singleton and Hamming bounds.

5.2.6 Example Consider the Hamming bound for the $(7, 4, 3)$ code C with generator matrix

$$G = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 \end{pmatrix}$$

which was discussed at the end of the Section 4.11. We have $n = 7$, $\delta = 3$. Thus

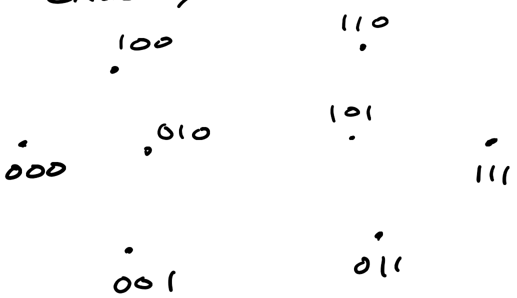
$$|C| \leq \frac{2^7}{\binom{7}{0} + \binom{7}{1}} = \frac{128}{8} = 16.$$

And in fact $|C| = 16$, so the bound is sharp for this code. This means $\mathbb{F}_2^7$ is exactly covered by the balls of radius 1 centred on the 16 words of C.

Similarly for $C = \{000, 111\}$ $\quad \delta = 3$ so $d = 1$

$$|C| = 2 = \frac{2^3}{\binom{3}{0}+\binom{3}{1}} = \frac{2^n}{\binom{n}{0}+\binom{n}{1}}$$

$\mathbb{F}_2^3$ is exactly covered by balls of radius 1.



5.3. The Griesmer Bound

5.3.1 Theorem [Griesmer Bound] Let C be a linear (n, k, δ) code. Then

$$n \geq \sum_{j=0}^{k-1} \left\lceil \frac{\delta}{2^j} \right\rceil. \tag{5.3.1}$$

Exercise

What is the lower bound on the length of a 2-error correcting code with 64 codewords given by the Griesmer bound?

5.3.1 Theorem [Griesmer Bound] Let C be a linear (n, k, δ) code. Then

$$n \geq \sum_{j=0}^{k-1} \left\lceil \frac{\delta}{2^j} \right\rceil . \tag{5.3.1}$$

Proof Let $L(k, \delta)$ denote the minimum length of a linear code of dimension k and distance δ.

Step ①: Prove that $L(k,\delta) \geq L(k-1, \lceil \frac{\delta}{2} \rceil) + \delta$.

Step ②: Use induction on k to prove that $n \geq \sum_{j=0}^{k-1} \lceil \frac{\delta}{2^j} \rceil$ (using the equation from Step ①)

Base case $k=1$ $\quad \sum_{j=0}^{0} \lceil \frac{\delta}{2^j} \rceil = \lceil \frac{\delta}{2^0} \rceil = \delta$

Clearly $n \geq \delta$ so base case is true.

Inductive hypothesis: Eq 5.3.1 holds for $k-1$, that is,

$$n \geq \sum_{j=0}^{k-2} \left\lceil \frac{\delta}{2^j} \right\rceil$$

Now prove Eq 5.3.1 for k.

$$\begin{aligned}
n &\geq L(k, \delta) && \text{Definition of } L(k,\delta) \\
&\geq L\big(k-1, \lceil \delta/2 \rceil\big) + \delta && \text{Equation 5.3.5} \\
&\geq \sum_{j=0}^{k-2} \left\lceil \frac{\lceil \delta/2 \rceil}{2^j} \right\rceil + \delta && \text{Inductive hypothesis } k-1 \\
&= \sum_{j=0}^{k-2} \left\lceil \frac{\delta}{2^{j+1}} \right\rceil + \delta && \text{Lemma 2.3.7} \\
&= \sum_{i=0}^{k-1} \left\lceil \frac{\delta}{2^i} \right\rceil && \text{Put } i = j+1.
\end{aligned}$$

Prove that $L(k,\delta) \geq L(k-1, \lceil \frac{\delta}{2} \rceil) + \delta$.

Fix k and δ. Let C be a linear code of dimension k and distance δ of minimal length $n = L(k, \delta)$. By Theorem 4.2.1 there is a word $c \in C$ of exact weight δ, and no smaller weight non-zero word. Since equivalent codes have the same parameters, wlog let $c = (\underbrace{11\cdots1}_{\delta} \mid \underbrace{00\cdots0}_{n-\delta}) = (\mathbf{1}_\delta \mid \mathbf{0}_{n-\delta})$,

Extend $\{c\}$ to a basis for C. Let G be a corresponding generator matrix for this basis, with first row c. Thus C has a generating matrix

$$\mathsf{G} = \left(\begin{array}{ccc|ccc} 1 & \cdots & 1 & 0 & \cdots & 0 \\ \hline * & * & * & & & \\ \vdots & \vdots & \vdots & & \mathsf{G}' & \end{array}\right).$$

First prove that the rows of G' are linearly independent.

$$\left(\begin{array}{c|c} 1 \cdots 1 & 0 \cdots 0 \\ \hline g_1 & g_1' \\ g_2 & g_2' \\ \vdots & \vdots \\ g_{k-1} & g_{k-1}' \end{array}\right)$$

Suppose the rows of G' are not linearly independent, so $\exists$ a subset of $\{g_1', g_2', \dots g_{k-1}'\}$ that sum to $\underset{\sim}{0}$.

For example, suppose $g_1' + g_2' + g_3' = \underset{\sim}{0}$.

Then $(g_1 | g_1') + (g_2 | g_2') + (g_3 | g_3') = (g_1 + g_2 + g_3 \mid 0 \dots 0)$.

Now $(g_1 + g_2 + g_3 \mid 0 \dots 0) \in C$ so $\| g_1 + g_2 + g_3 \| \leq \delta$.

Now consider the linear code $C' = f(C)$ generated by G'.

Let $w \in f(C)$ be a non-zero word of minimal weight. So there exists $u \in C$ with $u = (v \mid w)$ and $\|w\| = \delta'$. Then

$$\begin{array}{rcll}
\delta & \leq & \|u + c\| & u, c \in C,\ \ u \neq c \implies 0 \neq u + c \in C \\
& = & \|(v \mid w)\ +\ (\mathbf{1}_\delta \mid \mathbf{0}_{n-\delta})\| & \\
& = & \|(v + \mathbf{1}_\delta) \mid (w + \mathbf{0}_{n-\delta})\| & \\
& = & \|v + \mathbf{1}_\delta\|\ +\ \|w + \mathbf{0}_{n-\delta}\| & \\
& = & \delta\ -\ \|v\|\ +\ \|w\| & \text{Equation 2.2.2} \\
& \leq & \|u\| - \|v\|\ +\ \|w\| & 0 \neq u \in C \implies \|u\| \geq \delta \\
& = & \|v\| + \|w\| - \|v\|\ +\ \|w\| & u = (v \mid w) \\
& = & 2\|w\| = 2\delta'. &
\end{array}$$

5.1.2 Theorem [Singleton Bound] In any code (linear or not), $|C| \le 2^{n-\delta+1}$. In particular, if $|C| = 2^k$, then

$$n \ge k + \delta - 1. \tag{5.1.1}$$

5.2.3 Theorem [Hamming Bound] If C is a code of length n (linear or not) and distance δ then

$$|C| \le \frac{2^n}{\binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{d}} \qquad \text{where } d = \left\lfloor \frac{\delta - 1}{2} \right\rfloor.$$

5.3.1 Theorem [Griesmer Bound] Let C be a linear (n, k, δ) code. Then

$$n \ge \sum_{j=0}^{k-1} \left\lceil \frac{\delta}{2^j} \right\rceil. \tag{5.3.1}$$

Exercise: What do each of the Singleton, Hamming and Griesmer bounds tell us about the maximum number of codewords in a linear code with length $n=20$ and distance $\delta = 5$?

5.4.1 Theorem [Gilbert–Varshamov (G–V)] Let $n, k, \delta \in \mathbb{N}$, with $n > \delta \geq 2$. If

$$\binom{n-1}{0} + \binom{n-1}{1} + \cdots + \binom{n-1}{\delta-2} < 2^{n-k} \tag{5.4.1}$$

then there exists a linear (n, k, δ)-code.

Proof Suppose that the integers n, k and δ satisfy 5.4.1. We will construct an $n \times (n-k)$ matrix **H** with linearly independent columns and in which no set of $\delta - 1$ or fewer rows is linearly dependent. This matrix is a parity check matrix for a linear (n, k, d)-code for some $d \geq \delta$, by Theorem 4.9.1. Thus, by the Spoiling Lemma 5.1.1, there exists a linear (n, k, δ)-code.

Start building the matrix **H** by choosing the first $n-k$ rows of **H** to be the rows of the matrix $\mathbf{I}_{n-k}$. This guarantees that the columns of **H** are linearly independent. Furthermore, we are guaranteed that so far, we have $n-k$ rows, no set of $\delta - 1$ of which are linearly dependent.

$$H = \begin{pmatrix} I_{n-k} \\ \text{———} \\ \text{———} \\ \text{———} \end{pmatrix}_{n \times (n-k)}$$

Need
- linearly independent columns
- no set of $\delta - 1$ or fewer rows in linearly dependent.

Build H row by row, checking at each stage that no set of $\delta-1$ or fewer rows is lin. dep.
Given i rows, we can find a vector of length $n-k$ to put in row $i+1$ provided:

$$\binom{i}{0} + \binom{i}{1} + \binom{i}{2} + \cdots + \binom{i}{\delta-2} < 2^{n-k}.$$

To add the final row, we need

$$\binom{n-1}{0} + \binom{n-1}{1} + \cdots + \binom{n-1}{\delta-2} < 2^{n-k}.$$

Exercise:

(a) Show that the G-V bound guarantees the existence of a linear (8, 3, 4)-code.

(b) Use the construction technique of the proof of the G-V bound to find a parity check matrix for a linear (8, 3, 4)-code.

(c) From the p.c. matrix, find a corresponding generating matrix for your linear (8, 3, 4)-code.

5.4.2 Corollary If $n > \delta \geq 2$ then there exists a linear code C with length n and distance δ with

$$|C| \geq \frac{2^{n-1}}{\binom{n-1}{0} + \binom{n-1}{1} + \cdots + \binom{n-1}{\delta-2}}. \tag{5.4.3}$$

Proof The G–V bound guarantees the existence of a linear (n, k, δ)-code provided

$$\binom{n-1}{0} + \binom{n-1}{1} + \cdots + \binom{n-1}{\delta-2} < 2^{n-k}.$$

That is, if

$$2^k < \frac{2^n}{\binom{n-1}{0} + \binom{n-1}{1} + \cdots + \binom{n-1}{\delta-2}}. \tag{5.4.4}$$

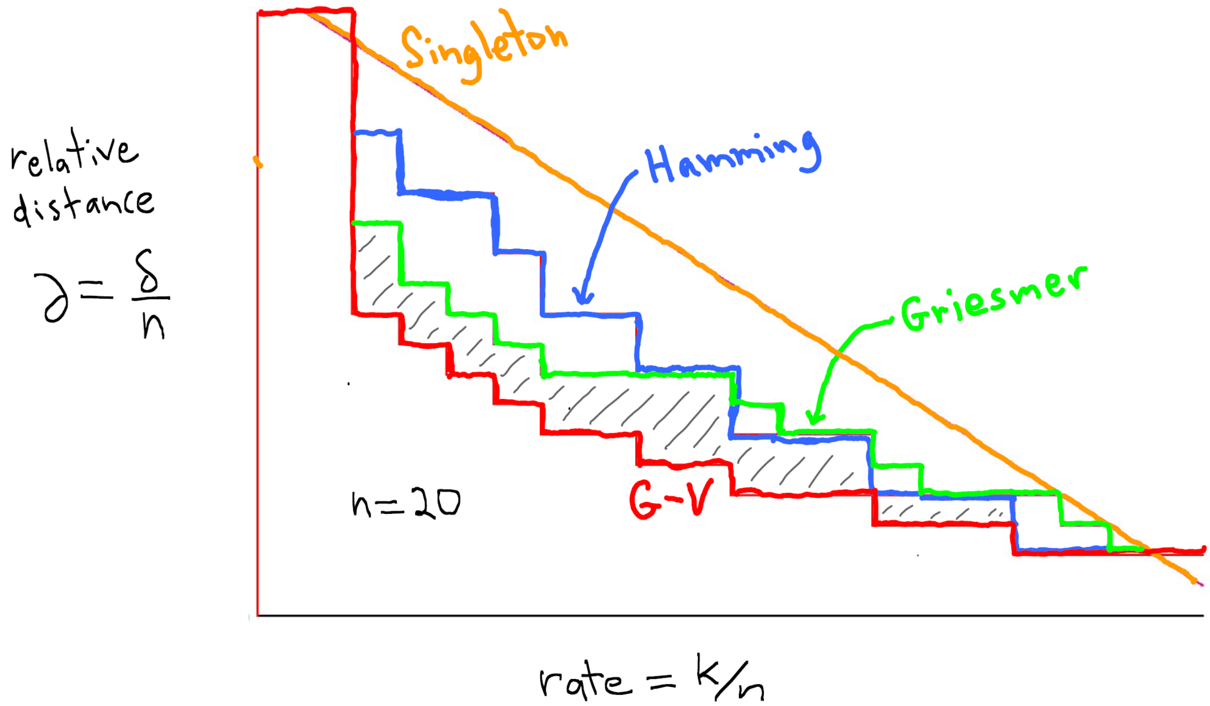
Let k' be the largest value of k for which inequality 5.4.4 is satisfied, so (by G–V) there does exist a linear (n, k', δ) code, C. We show that C satisfies inequality 5.4.3.

Inequality 5.4.4 is not satisfied for $k' + 1$, so

$$2^{k'+1} \geq \frac{2^n}{\binom{n-1}{0} + \binom{n-1}{1} + \cdots + \binom{n-1}{\delta-2}}$$

$$\therefore \qquad |C| = 2^{k'} \geq \frac{2^{n-1}}{\binom{n-1}{0} + \binom{n-1}{1} + \cdots + \binom{n-1}{\delta-2}}. \qquad \square$$

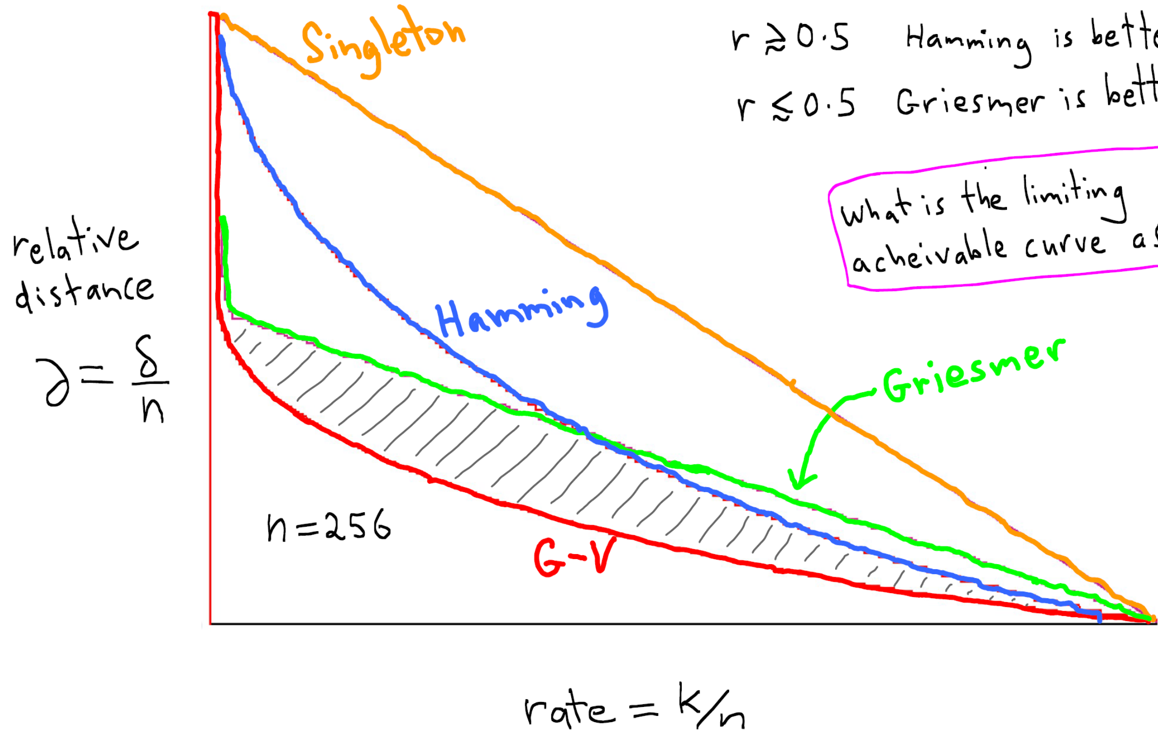
<u>Exercise</u>: What does the G-V. bound tell us about the number of codewords in a linear code of length $n = 20$ and distance $\delta = 5$?



$r \gtrsim 0.5$ Hamming is better bound.

$r \lesssim 0.5$ Griesmer is better bound.

What is the limiting acheivable curve as $n \to \infty$?



5.5. Perfect Codes

5.5.1 Definition A code C of length n and odd distance $\delta = 2t+1$ is called a *perfect code* if C attains the Hamming bound given in Theorem 5.2.3. That is, C is perfect iff

$$|C| = \frac{2^n}{\binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{t}}.$$

5.5.2 Theorem If C is a perfect code of length n and distance $\delta = 2t+1$ then C will correct all error patterns of weight less than or equal to t, <u>and no other error patterns.</u> Every word in $\mathbb{F}_2^n$ lies in exactly one closed ball of radius t centred on a codeword.

Every word of length n is at distance $\leq t$ from exactly one codeword.

<u>Exercise</u>: Find all possible parameters (n, k, δ) for perfect codes of lengths 5, 6, 7, 8.

When is $\binom{n}{0}+\binom{n}{1}+\cdots+\binom{n}{t}$ a power of 2?

① When $t=0$. Here $\delta=1$ and we have the $(n,n,1)$-code consisting of all words of length n.

② When $t=\frac{n-1}{2}$ and n is odd.
Here $\delta=n$ and we have the $(n,1,n)$-code
$C=\{\underset{\sim}{0},\underset{\sim}{1}\}$.

③ When $t=1$ and $n+1$ is a power of 2.
Here $\delta=3$. $\binom{n}{0}+\binom{n}{1}=1+n=2^r$ for some $r\in\mathbb{Z}^+$.

$$|C|=\frac{2^n}{\binom{n}{0}+\binom{n}{1}}=2^{n-r}=2^{2^r-1-r}.$$

The "Hamming codes", a family of $(2^r-1,\ 2^r-1-r,\ 3)$-codes.

④ When $t=5$ and $n=90$.

$$\binom{90}{0}+\binom{90}{1}+\binom{90}{2}=2^{12} \qquad |C|=\frac{2^{90}}{2^{12}}=2^{78}$$

But no $(90,78,5)$-code exists!

⑤ When $t=7$ and $n=23$.

$$\binom{23}{0}+\binom{23}{1}+\binom{23}{2}+\binom{23}{3}=2048=2^{11}.$$

$$|C|=\frac{2^{23}}{2^{11}}=2^{12}$$

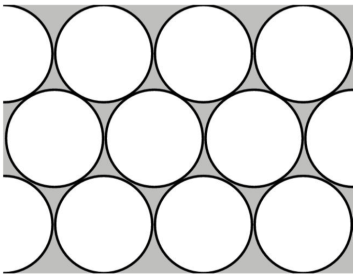
The Golay code is a perfect $(23,12,7)$-code.

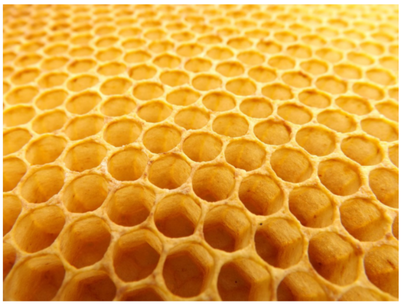
5.6. Interlude: Sphere Packing

A perfect code has the property that it partitions $\mathbb{F}_2^n$ precisely into identical non-overlapping spheres (really, balls). In this case 100% of the elements of $\mathbb{F}_2^n$ are covered by spheres. Clearly(?) this is impossible in $\mathbb{R}^n$ ($n \geq 2$).

5.6.1 Question Let $n \geq 2$. What is the limiting maximal density of space that can be covered by non-overlapping unit (hyper)-balls in $\mathbb{R}^n$?

(To make this rigorous one considers any packing of spheres, calculates the percentage of volume of a cube of side length s that is occupied by the packing, and considers the limit as $s \to \infty$.)





Honeycomb

Hexagonal close packing with Hales

n	Optimal Density	Structure	Proof
2	$\frac{\pi}{2\sqrt{3}} \simeq 0.9069$	Hexagonal lattice (honeycomb)	1940, Toth
3	$\frac{\pi}{3\sqrt{2}} \simeq 0.7405$	Hexagonal close packed (cannon balls)	(Conjectured by Kepler 1611) 1998, Hales
≥ 4	?	Unknown before 2016	
8	$\frac{\pi^4}{384} \simeq 0.2537$	E_8 lattice	2016, Viazovska
24	$\frac{\pi^{12}}{12!} \simeq 0.001930$	Leech lattice	2016, Viazovska et al

The existence of the Leech lattice in some way "explains" the existence of the unique perfect code with $\delta > 3$, namely the $(23, 12, 7)$ code that we shall construct later.